Professional Industry Defense Speaker Notes

Author: Sukh Sandhu

Presentation Plan

The defense is planned for 15 minutes and follows the required sections: industry context, integrated system overview, prior project integration, technical decisions and tradeoffs, ethical considerations, evaluation, limitations, professional relevance, and next steps.

Mentor Q&A Preparation

Likely questions include why healthcare operations was selected, why patient-facing decisions require human approval, why public benchmark datasets were used, how the system would be validated, and what failure cases matter most. Answers should emphasize reproducibility, bounded scope, transparency, and the need for domain validation.

Slide Timing

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Total planned time: 15 minutes.

Slide 1 - Opening (1 minute)
Introduce the healthcare operations context and state that the system is a controlled triage workbench, not a diagnostic tool.

Slide 2 - Industry Context (1.5 minutes)
Explain the real workflow problem: incomplete structured data, model signals, scanned artifacts, and the need for accountable human review.

Slide 3 - Integrated System Overview (1.5 minutes)
Walk through the data-quality layer, statistical checks, supervised ML, CNN visual classifier, VAE quality review, and agentic routing layer.

Slide 4 - Prior Project Integration (2 minutes)
Identify at least three prior projects and what each contributed: data workflow, statistical reasoning, supervised ML, deep learning, generative AI, and agentic workflow.

Slide 5 - Technical Decisions (2 minutes)
Defend why the system uses interpretable routing, stratified evaluation, public benchmark datasets for reproducibility, and a hard human-approval boundary.

Slide 6 - Evaluation (1.5 minutes)
Summarize scenario results and explain how the system reacts to low quality, low confidence, visual uncertainty, generative outlier flags, and patient-facing status.

Slide 7 - Ethics and Safeguards (2 minutes)
Discuss automation bias, unequal performance, accountability, transparency, and why patient-facing decisions stay with humans.

Slide 8 - Limitations and Failure Cases (1.5 minutes)
Be direct about prototype signals, benchmark datasets, lack of clinical validation, and the need for domain data before deployment.

Slide 9 - Professional Relevance (1 minute)
Connect the work to real AI roles: reproducible analysis, modeling, evaluation, responsible AI, and stakeholder communication.

Slide 10 - Closing and Defense Readiness (1 minute)
Close with next steps: validation, monitoring, fairness checks, threshold review, and human feedback loops.